

TUGAS SISTEM MULTIMEDIA

Image Processing Pipeline dan JPEG Compression



Disusun Oleh:

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Program Studi S1 Informatika

1. Tujuan Praktikum

Praktikum ini bertujuan untuk memahami proses pengolahan citra digital menggunakan Python, meliputi konversi grayscale, bitplane slicing, split RGB, dan kompresi JPEG.

2. Tools dan Library

- Visual Studio Code
- Python 3.11
- Pillow
- NumPy
- OpenCV
- Matplotlib

3. Source Code dan Implementasi

Program dibuat menggunakan Python untuk melakukan manipulasi citra digital dan kompresi JPEG dengan beberapa quality factor.

- Source Code

```
from PIL import Image
import numpy as np
import os

# Load gambar
img = Image.open("gambar.jpg")

print("Ukuran:", img.size)
print("Mode:", img.mode)

# Convert ke numpy
img_np = np.array(img)

# Ambil channel RGB
R = img_np[:, :, 0]
G = img_np[:, :, 1]
B = img_np[:, :, 2]

# Convert grayscale
gray = (0.299 * R + 0.587 * G + 0.114 * B).astype(np.uint8)

# Simpan grayscale
gray_img = Image.fromarray(gray)
gray_img.save("grayscale.png")
```

```

print("Grayscale selesai")

# Bitplane
for i in range(8):
    bitplane = ((gray >> i) & 1) * 255
    Image.fromarray(bitplane.astype(np.uint8)).save(f"bitplane_{i}.png")

print("Bitplane selesai")

# Split RGB
zeros = np.zeros_like(R)

red_img = np.stack([R, zeros, zeros], axis=2)
green_img = np.stack([zeros, G, zeros], axis=2)
blue_img = np.stack([zeros, zeros, B], axis=2)

Image.fromarray(red_img).save("red_channel.png")
Image.fromarray(green_img).save("green_channel.png")
Image.fromarray(blue_img).save("blue_channel.png")

print("Split RGB selesai")

# Kompresi JPEG
qualities = [10, 50, 90]

for q in qualities:
    filename = f"compressed_Q{q}.jpg"

    img.save(filename, "JPEG", quality=q)

    size_kb = os.path.getsize(filename) / 1024

    print(f"Q={q} -> {size_kb:.2f} KB")

# Analisis raw size
width, height = img.size

channels = 3
bit_depth = 8

raw_size = width * height * channels * bit_depth / 8

print(f"Raw size: {raw_size / 1024:.2f} KB")

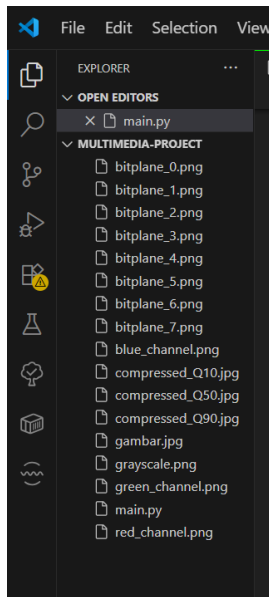
```

- Screenshot Source Code

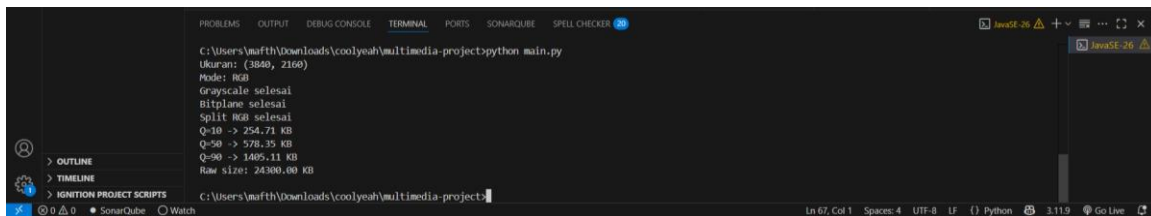
4. Hasil Praktikum

- Konversi grayscale berhasil
- Bitplane slicing berhasil
- Split RGB berhasil
- Kompresi JPEG berhasil

- Screenshot hasil file



5. Hasil Terminal



Ukuran: (3840, 2160)

Mode: RGB

Grayscale selesai

Bitplane selesai

Split RGB selesai

Q=10 -> 254.71 KB

Q=50 -> 578.35 KB

Q=90 -> 1405.11 KB

Raw size: 24300.00 KB

6. Analisis

Quality	Ukuran
Q10	254.71 KB
Q50	578.35 KB
Q90	1405.11 KB

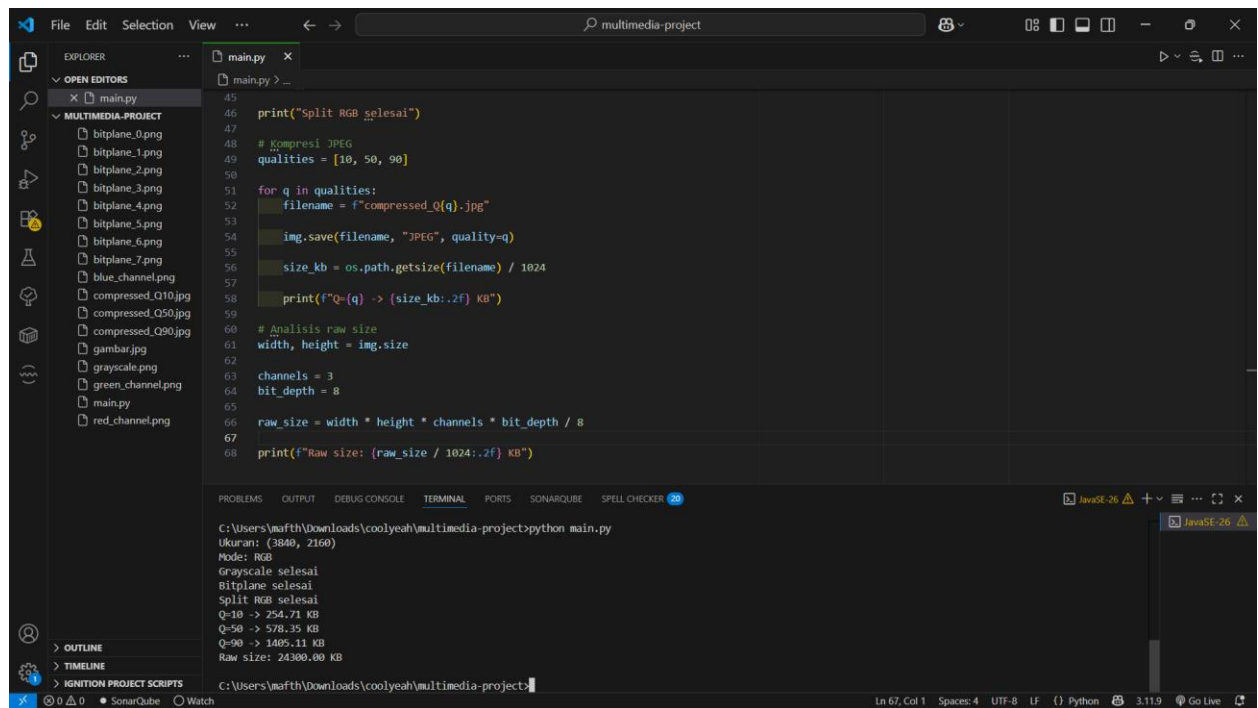
Hasil menunjukkan bahwa semakin kecil quality factor, maka ukuran file menjadi lebih kecil namun kualitas gambar menurun. Sebaliknya, quality factor tinggi menghasilkan kualitas gambar lebih baik dengan ukuran file lebih besar.

7. Kesimpulan

Praktikum berhasil dijalankan menggunakan Python dan Visual Studio Code. Teknik grayscale, split RGB, bitplane slicing, dan kompresi JPEG berhasil diterapkan dengan baik. Kompresi JPEG terbukti efektif mengurangi ukuran file gambar.

8. Screenshot Implementasi

Berikut tampilan full Source Code dan hasil terminal



The screenshot displays the Visual Studio Code interface with a Python script named `main.py` open in the editor. The script performs several operations: it prints a completion message, compresses an image using JPEG with quality factors of 10, 50, and 90, calculates the size of the compressed files, and prints the raw size of the original image. The terminal at the bottom shows the output of running the script, including the image dimensions (3840x2160), mode (RGB), and the resulting file sizes for each quality factor and the raw image size.

```
45 print("Split RGB selesai")
46
47 # Kompresi JPEG
48 qualities = [10, 50, 90]
49
50 for q in qualities:
51     filename = f"compressed_Q{q}.jpg"
52     img.save(filename, "JPEG", quality=q)
53     size_kb = os.path.getsize(filename) / 1024
54     print(f"Q-{q} -> {size_kb:.2f} KB")
55
56 # Analisis raw size
57 width, height = img.size
58 channels = 3
59 bit_depth = 8
60
61 raw_size = width * height * channels * bit_depth / 8
62
63 print(f"Raw size: {raw_size / 1024:.2f} KB")
```

Terminal Output:

```
C:\Users\wafth\Downloads\coolyeah\multimedia-project>python main.py
Ukuran: (3840, 2160)
Mode: RGB
Grayscale selesai
Bitplane selesai
Split RGB selesai
Q-10 -> 254.71 KB
Q-50 -> 578.35 KB
Q-90 -> 1485.11 KB
Raw size: 24300.00 KB
```